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UID:-22BCS10233

SEC:-636/A

SUBJECT

1. Write a Java program to calculate the square root of a number entered by the user. Use try-catch to handle invalid inputs (e.g., negative numbers or non-numeric values).

-> import java.util.Scanner;

```
public class Main {  
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);  
        try {  
            System.out.print("Enter a number: ");  
            double num = scanner.nextDouble();  
            if (num < 0) {  
                throw new IllegalArgumentException("Square root of a negative number is not real.");  
            }  
            double result = Math.sqrt(num);  
            System.out.println("Square root: " + result);  
        } catch (IllegalArgumentException e) {  
            System.out.println("Error: " + e.getMessage());  
        } catch (Exception e) {  
            System.out.println("Invalid input! Please enter a valid numeric value.");  
        } finally {  
            scanner.close();  
        }  
    }  
}
```

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OUTPUT



```
Output
Enter a number: 64
Square root: 8.0

=== Code Execution Successful ===
```

2. Write a Java program to simulate an ATM withdrawal system. The program should:

Ask the user to enter their PIN.

Allow withdrawal if the PIN is correct and the balance is sufficient.

Throw exceptions for invalid PIN or insufficient balance.

Ensure the system always shows the remaining balance, even if an exception occurs.

-> import java.util.Scanner;

```
class InvalidPINException extends Exception {
    public InvalidPINException(String message) {
        super(message);
    }
}

class InsufficientBalanceException extends Exception {
    public InsufficientBalanceException(String message) {
        super(message);
    }
}

public class Main {
    private static final int CORRECT_PIN = 1234;
    private static double balance = 5000.0; // Initial balance
```

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```
public static void main(String[] args) {  
    Scanner scanner = new Scanner(System.in);  
    try {  
        System.out.print("Enter your PIN: ");  
        int enteredPIN = scanner.nextInt();  
  
        if (enteredPIN != CORRECT_PIN) {  
            throw new InvalidPINException("Incorrect PIN. Please try again.");  
        }  
        System.out.print("Enter amount to withdraw: ");  
        double withdrawAmount = scanner.nextDouble();  
        if (withdrawAmount > balance) {  
            throw new InsufficientBalanceException("Insufficient balance for this withdrawal.");  
        }  
        balance -= withdrawAmount;  
        System.out.println("Withdrawal successful! You withdrew: $" + withdrawAmount);  
    } catch (InvalidPINException | InsufficientBalanceException e) {  
        System.out.println("Error: " + e.getMessage());  
    } catch (Exception e) {  
        System.out.println("Invalid input! Please enter a valid number.");  
    } finally {  
        System.out.println("Remaining balance: $" + balance);  
        scanner.close();  
    }  
}
```

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OUTPUT

```
Output Clear  
Enter your PIN: 1234  
Enter amount to withdraw: 2000  
Withdrawal successful! You withdrew: $2000.0  
Remaining balance: $3000.0  
  
=== Code Execution Successful ===
```